

PALLETPILOT · DESIGN DECISION RECORD

PalletPilot recommendations accepted

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DOCUMENT CONTROL

Document ID	PLP-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/palletpilot		
Source	docs/decisions/0002-recommendations-accepted.md @ fe1c41f		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of the TRL 3 review recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted for items O4 to O6; items O1 to O3 remain proposed

■ Context

The TRL 3 review note (docs/REVIEW.md , session 2026-09-25, TRL 3) and PLP-DDR-001 left six items as "Proposed, awaiting Amish". Three of them (O4, O5, O6) carried a recommendation; three (O1, O2, O3) did not. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item with a recommendation is therefore decided in favor of it, and where the recommendation named one of several options, that option is the decision. Items with no recommendation stay open. TRL 4 remains on hold by Amish's instruction, so any recommendation that needs building, testing or purchasing is recorded as decided but on hold.

■ Options considered

The options for each item are those in docs/REVIEW.md (TRL 3 session), PLP-DDR-001 Table 2 and PLP-CAL-001 v0.1.

■ Decision

Table 1. Items decided on 2026-09-25.

#	Item	Decision	What changed in the repo
O4	Bumper-only stopping (R8): (a) limit bumper-only speed to 0.15 m/s, or (b) an edge with 65 mm or more of travel	Decided by Amish, 2026-09-25: go with recommendation. Option (a): any operation that relies on the bumper alone is limited to 0.15 m/s. Because the lidar layer is untested, follow mode also runs at 0.15 m/s or less until that layer is built and tested (was 0.2 m/s)	R8 target restated from 0.2 to 0.15 m/s and now met on paper (38 mm stop within 40 mm of travel); R4 interim follow limit 0.2 to 0.15 m/s; PLP-CAL-001 stop table and [S3] line; safety sections in PLP-PRC-001 and README; note on PLP-DWG-001
O5	Mass (1.7 kg) and cost (\$30) overruns, and the R5 bearing gap	Decided by Amish, 2026-09-25: go with recommendation. Accept both overruns for now and recheck them at a motor quote, with no budget change. For R5, evaluate angle-of-arrival UWB and lidar leg tracking at TRL 4; decided but on hold, since TRL 4 is on hold	budget_usd stays at 1550. R15 mass and R17 are still reported as not met, marked "accepted for now". No R5 design change
O6	Stop chain for PL d (R9): a second contactor or a driver with rated safe torque off	Decided by Amish, 2026-09-25: go with recommendation. Add a second main contactor in series, one per safety relay channel	BOM line 6 from \$45 to \$75; model part 6 now shows two contactors; kit mass 41.7 to 42.0 kg; BOM total \$1,580 to \$1,610; R9 target text restated; PLP-DWG-001 to Rev P2; PLP-CAL-001 to v0.2. R9 stays at risk until the PL is calculated

Table 2. Items still open.

#	Item	Status
O1	Named site and co-design partners (partner types were decided in PLP-DDR-001 D9)	Proposed, awaiting Amish; partners are picked per area later
O2	Classification of follow mode under ISO 3691-4 and the safety functions it then needs	Proposed, awaiting Amish; no recommendation made
O3	Which donor jack models and steering yokes to support first	Proposed, awaiting Amish; no recommendation made

■ Consequences

- Requirement status (PLP-CAL-001 v0.2): 10 met, 3 not met (R5, R15 mass, R17), 2 at risk (R7, R9), 3 not verifiable at TRL 3 (R1, R6, R16). Before these decisions it was 9 met, 3 not met, 3 at risk and 3 not verifiable.
- Numbers that moved: kit mass 41.7 to 42.0 kg; BOM total \$1,580 to \$1,610 (3.9 % over \$1,550); energy from the pack 292 to 293 Wh per shift; unpowered push force rise 3.9 % to 4.0 %; bumper-only speed limit 0.2 to 0.15 m/s.
- Documents revised: PLP-REQ-001 and PLP-PRC-001 to v0.4, PLP-CAL-001 to v0.2, PLP-DDR-001 to v0.2, PLP-DWG-001 to Rev P2. `cad/src/model.py` re-exported to STEP and STL; concept media regenerated. `project.yaml` is unchanged apart from listing this record as evidence; the pitch, problem and budget are unchanged.
- Cross-repo actions: none. PalletPilot uses its own 24 V pack and shares no interface with another repo.
- On hold (TRL 4): the R5 evaluation of angle-of-arrival UWB and lidar leg tracking, which needs bench measurement. The motor quote that rechecks the overruns and the PL calculation of the stop chain are paper work at TRL 3 and are listed as the next step; no part is to be bought.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building, testing or buying.